Computing Koopman Modes

Consider a discrete time system:

(DS1) $$x_{k+1} = \underbrace{F(x_k)}_{\mathbb{R}^d \to \mathbb{R}^d}, \quad k = 0, 1, 2, \ldots$$

The Koopman operator acts on observables $g : \mathbb{R}^d \to \mathbb{C}$,

$$[\mathcal{K} g](x) = [g \circ F](x)$$

Key Idea: Study the linear dynamics of observables,

$$g_{k+1} = \mathcal{K} g_k, \quad k = 0, 1, 2, \ldots$$

~~[illegible]~~ ⇒ Eigenfunctions of $\mathcal{K}$ encode state-space geometry. ~~e.g., invariant sets.~~

⇒ Level sets provide information about invariant sets, trapping regions, basins of attraction, and linear coordinates for whole basin of attraction.

⇒ See Example: Van der Pol (Page 2)

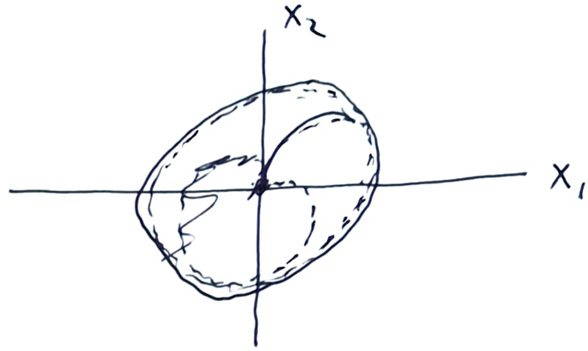
Today / Q: How do compute eigenfunctions of $\mathcal{K}$ from trajectory data?

Example: Isochrons of Van der Pol Oscillator

⇒ ~~Simple model w/stable limit~~ cycle

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = \mu(1-x_1)^2 x_2 - x_1$$



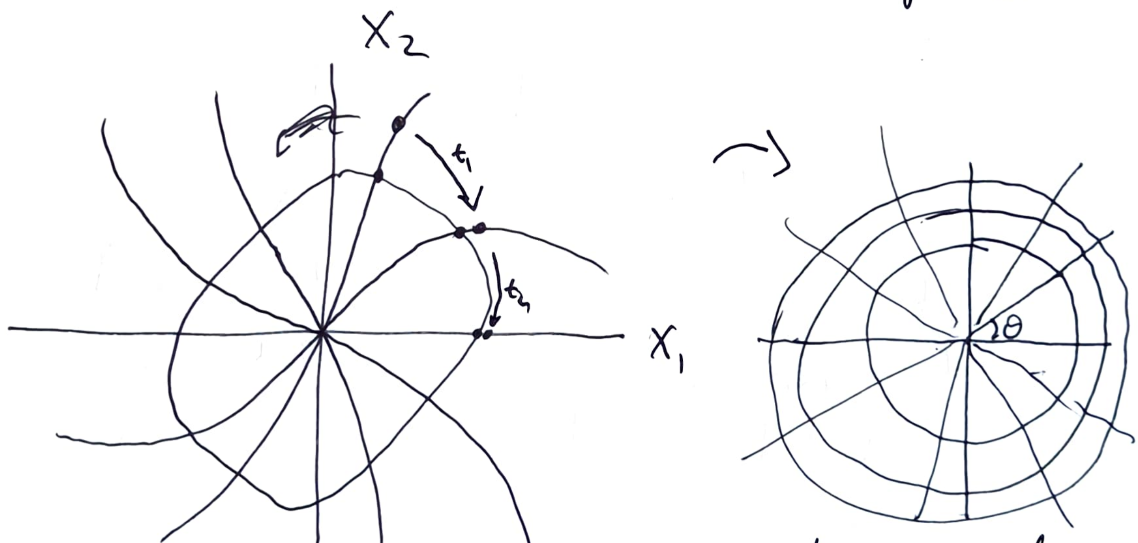
$\mu = 0.3$ $(\omega_0 \approx 0.995)$

⇒ Simple 2D model w/stable limit ~~c~~ cycle, origin unstable

⇒ Eigenvalues of Koopman at ~~[illegible]~~ $\exp(ik\omega_0 \Delta t)$ for $k = 0, \pm 1, \pm 2, \ldots$

⇒ Level sets of these eigenfunctions are <u>isochrons</u>, ~~all~~

~~Any~~ All initial conditions along an isochron will converge to ~~con~~ an orbit on the attractor w/the same phase



Linear periodic dynamics in isochron coords

⇒ ~~Together w/isostables~~, Used for phase reduction of oscillators

⇒ ~~Taken w/isostables~~, provide global linearization

The Power Method

Suppose that $g = c_1 v_1 + \cdots + c_n v_n$, where

$$K v_j = \lambda_j v_j, \quad \text{for } j = 1, \dots, n,$$

and $|\lambda_1| > |\lambda_2| \geq \dots \geq |\lambda_n| \geq 0$. ~~We can ex approximate~~

The power method from NLA allows us to extract the "dominant" eigenfunction of K from trajectory data:

$$K^m g = c_1 \lambda_1^m v_1 + \cdots + c_n \lambda_n^m v_n$$

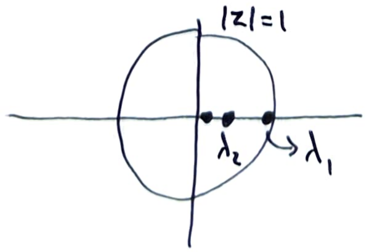
$$= \lambda_1^m \Big(c_1 v_1 + \underbrace{\left(\frac{\lambda_2}{\lambda_1}\right)^m}_{\to 0} c_2 v_2 + \cdots + \underbrace{\left(\frac{\lambda_n}{\lambda_1}\right)^m}_{\to 0} c_n v_n \Big)$$

$$\approx \lambda_1^m c_1 v_1 \qquad \text{as } m \to \infty$$

$\Rightarrow$ $\dfrac{K^m g}{\|K^m g\|}$ aligns w/the dominant eigenvector v_1 as $m \to \infty$.

$\Rightarrow$ Convergence can be slow when $|\lambda_2| / |\lambda_1| \approx 1$.

$\Rightarrow$ ~~Often effective~~ Can be effective for systems with "resonance"



"Fourier" "Laplace/Birkhoff" Averages

Q: What if eigenvalues ~~are~~ are all on $\mathbb{T} = \{|z|=1\}$?
and say, $\lambda_1 = 1$.

~~Idea: Target~~

If $|\lambda_1| = |\lambda_2| = \cdots = |\lambda_n| = 1$, then the expansion

$$K^m g = c_1 \lambda_1^m v_1 + \cdots + c_n \lambda_n^m v_n$$

does not converge as $m \to \infty$, since each term ~~cycles~~ remains constant in magnitude and "cycles" around unit circle.

Idea: "Average" out the cycling phases and target $\lambda_1 = 1$.

Define

$$\begin{aligned} w_m &= \frac{1}{m}\left(g + Kg + \cdots + K^{m-1} g\right) \\ &= \frac{1}{m} c_1 \left(1 + \lambda_1 + \cdots + \lambda_1^{m-1}\right) v_1 \\ &+ \frac{1}{m} c_2 \left(1 + \lambda_2 + \cdots + \lambda_2^{m-1}\right) v_2 \\ &\;\;\vdots \\ &+ \frac{1}{m} c_n \left(1 + \lambda_n + \cdots + \lambda_n^{m-1}\right) v_n \end{aligned}$$

The question is, how does the series ~~of partial~~ behave?

$$\lim_{m \to \infty} \frac{1}{m} \sum_{j=0}^{m-1} \lambda^j \qquad \text{for } \lambda \in \{\lambda_1, \ldots, \lambda_n\}.$$